

title

Observación 1. Sean $x, y \in \mathbb{R}^n$, tomamos $\forall k \in \mathbb{N}_n : a_k := |x_k - y_k| \geq 0$. Entonces,

$$\max_{1 \leq k \leq n} a_k = \left(\max_{1 \leq k \leq n} a_k^p \right)^{1/p} \leq \left(\sum_{k=1}^n a_k^p \right)^{1/p} \leq \left(n \max_{1 \leq k \leq n} a_k^p \right)^{1/p} = n^{1/p} \max_{1 \leq k \leq n} a_k.$$

Es decir, $d_\infty(x, y) \leq d_p(x, y) \leq n^{1/p} d_\infty(x, y)$, así que $\lim_{p \rightarrow \infty} d_p(x, y) = d_\infty(x, y)$.